

Coding & Solution

Date	04 JULY2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of the implemented code and the overall solution. The submission demonstrates working functionality, clean code practices, and alignment with the defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/geethikakonduru9/OptiCrop-Smart-Agricultural-Production-Optimization-Engine
Programming Language(s)	Python 3.10+, HTML5, CSS3, JavaScript
Framework(s) Used	Flask (Backend), Bootstrap 5 (Frontend), SQLite Database
Key Features Implemented	• Machine Learning-based Crop Recommendation using Random Forest Classifier • User Registration and Secure Login • SQLite Database Integration • Data Cleaning (Missing Values & Duplicate Removal) • Exploratory Data Analysis (Univariate, Bivariate & Multivariate Analysis) • Prediction Model stored as crop_model.pkl • Flask-based Web Application with Responsive User Interface • Farmer-friendly Crop Recommendation System
Pending / Incomplete Features	• Weather API Integration • Soil Sensor Integration • Cloud Deployment (AWS/Render/Heroku) • Mobile Application Support • Regional Language Support

Setup / Run Instructions	1. Clone the repository. 2. Install dependencies using <code>pip install -r requirements.txt</code> . 3. Ensure <code>database.db</code> and <code>crop_model.pkl</code> are in the project folder. 4. Run the application using <code>python app.py</code> . 5. Open <code>http://127.0.0.1:5000</code> in a web browser.
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Code Quality Checklist

S.No	Criteria	Status
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The **OptiCrop: Smart Agricultural Production Optimization Engine** project is developed using **Python, Flask, SQLite, HTML, CSS, and Machine Learning**. The system performs crop prediction based on agricultural parameters using a trained **Random Forest** model. The project includes user authentication, database management, exploratory data analysis, and a responsive web interface. The source code is maintained in the GitHub repository and follows a modular structure for improved readability, maintainability, and future enhancements.